

mars [12] and NGS [21]. Neural Grammars have been introduced in the natural language community as an effective strategy to learn PCFGs. They are neural parameterizations of PCFG and it is possible to learn the structure of the grammar by enumerating a set of candidate rules and using neural networks to learn their probabilities. Differently from DeepStochLog, they are restricted to context-free grammars. Furthermore, Neural Grammars [12] do not consider subsymbolic inputs (as in all our tasks T1-T6). Different from the probabilistic interface of DeepStochLog, NGS uses backsearch, a greedy search that defines the backward feedback from the grammar to the neural nets. While this makes NGS very scalable, the backsearch must be defined *per-program*, while DeepStochLog backpropagates evidence automatically through any NDCG. Neural Attribute Grammars [27] integrate attribute grammars with neural networks. While this is also an expressive grammatical framework, they are quite different from DeepStochLog in their approaches and applications, and also are not applied to subsymbolic data.

Third, many systems in the neural symbolic community [11, 10, 37] obtain differentiable logics by relaxing logical programs or theories using fuzzy logic and t-norms. While the shift in semantics from probabilistic to fuzzy logic has known issues [41], fuzzy logic allows for more scalable systems as compared to probabilistic logic based on the possible world semantics. But by exploiting the stochastic grammars, DeepStochLog shows the same benefits as fuzzy logic in terms of computational complexity (i.e. no disjoint-sum problem required) by resorting to an alternative probabilistic semantics.

8 Conclusions

We have introduced a novel and very expressive neural symbolic model based on stochastic logic programming, that allows to integrate symbolic knowledge with subsymbolic representations, that scales well, and gives state-of-the-art results on various neural symbolic computation tasks.

There are several limitations of DeepStochLog that we want to explore in further research. First, DeepStochLog does not yet learn the structure of the rules, while the neural theorem prover [31], DiffLog [36] and the neural grammars [12] can all enumerate rules and then identify the most relevant ones. Second, DeepStochLog’s inference could be further optimised by parallelization of the circuit using ideas from TensorLog [5]. Third, SLPs and hence, DeepStochLog, may lose probability mass due to failing derivations. This can be addressed by normalizing and computing the partition function [6]. It would be interesting to approximate the partition function and also to further speed up the inference by sampling or by searching for the k-best derivations. Finally, it would be interesting to explore the use of DeepStochLog as a generative model to generate sequences.

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